THE COLIMIT OF THE STRUCTURE SHEAF ALONG THE FROBENIUS IS A SHEAF FOR THE H TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVK (sources/stacks-flat/pinned-original-source.tex line 11855).

Allowed standard prerequisites (editorial).

- Characterisation of h sheaves by four conditions. Same chapter, Lemma labelled lemma-refine-check-h, cited at proof lines 11875-11877. Reduces the sheaf property to the four conditions checked in the proof.
- Sections of a directed colimit and the fppf sheaf property of the structure sheaf. Sheaves, Lemma labelled sheaves-lemma-directed-colimits-sections, cited at proof lines 11870-11872; Descent, Lemma labelled descent-lemma-sheaf-condition-holds, cited at proof lines 11879-11881. Identifies the sections on quasi-compact and quasi-separated objects and settles condition (2).
- The two generators examples and the funny blowup lemma. Same chapter, Examples labelled example-one-generator and example-two-generators, invoked at proof lines 11886-11888 and 11912-11913; Lemma labelled lemma-funny-blow-up, cited at proof lines 11912-11916. Supplies the explicit squares used in conditions (3) and (4).

Source reference. Original definition 0EU1: sources/stacks-flat/pinned-original-source.tex, source lines 10504–10504. Same-chapter pointer into the author environment definition at source line 10504; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 009F: sources/stacks-chow/tags.txt, source lines 351–351. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11873 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0EVI: sources/stacks-flat/pinned-original-source.tex, source lines 11501–11501. Same-chapter pointer into the author environment lemma at source line 11501; invoked at proof line 11877 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 03DT: sources/stacks-chow/tags.txt, source lines 4432–4432. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11881 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0EVG: sources/stacks-flat/pinned-original-source.tex, source lines 11474–11474. Same-chapter pointer into the author environment example at source line 11474; invoked at proof line 11888 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0EVH: sources/stacks-flat/pinned-original-source.tex, source lines 11488–11488. Same-chapter pointer into the author environment example at source line 11488; invoked at proof line 11910 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVJ: sources/stacks-flat/pinned-original-source.tex, source lines 11782–11782. Same-chapter pointer into the author environment lemma at source line 11782; invoked at proof line 11911 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let p be a prime number. Let S be a scheme over $\mathbf{F}_p$. Let $(Sch/S)_h$ be a site as in Definition 0EU1. There is a unique sheaf $\mathcal{F}$ on $(Sch/S)_h$ such that*

$$\mathcal{F}(X) = \operatorname{colim}_F \Gamma(X, \mathcal{O}_X)$$

for any quasi-compact and quasi-separated object X of $(Sch/S)_h$.

Proof. Denote $\mathcal{F}$ the Zariski sheafification of the functor

$$X \longrightarrow \operatorname{colim}_F \Gamma(X, \mathcal{O}_X)$$

For quasi-compact and quasi-separated schemes X we have $\mathcal{F}(X) = \operatorname{colim}_F \Gamma(X, \mathcal{O}_X)$. by Sheaves, Lemma 009F and the fact that $\mathcal{O}$ is a sheaf for the Zariski topology. Thus it suffices to show that $\mathcal{F}$ is a h sheaf. To prove this we check conditions (1), (2), (3), and (4) of Lemma 0EVI. Condition (1) holds because we performed an (almost unnecessary) Zariski sheafification. Condition (2) holds because $\mathcal{O}$ is an fppf sheaf (Descent, Lemma 03DT) and if A is the equalizer of two maps $B \to C$ of $\mathbf{F}_p$-algebras, then $\operatorname{colim}_F A$ is the equalizer of the two maps $\operatorname{colim}_F B \to \operatorname{colim}_F C$.

We check condition (3). Let A, f, J be as in Example 0EVG. We have to show that

$$\operatorname{colim}_F A = \operatorname{colim}_F A/J \times_{\operatorname{colim}_F A/fA+J} \operatorname{colim}_F A/fA$$

This reduces to the following algebra question: suppose $a', a'' \in A$ are such that $F^n(a' - a'') \in fA + J$. Find $a \in A$ and $m \geq 0$ such that $a - F^m(a') \in J$ and $a - F^m(a'') \in fA$ and show that the pair (a, m) is uniquely determined up to a replacement of the form $(a, m) \mapsto (F(a), m+1)$. To do this just write $F^n(a'-a'') = fh+g$ with $h \in A$ and $g \in J$ and set $a = F^n(a') - g = F^n(a'') + fh$ and set $m = n$. To see uniqueness, suppose (a_1, m_1) is a second solution. By a replacement of the form given above we may assume $m = m_1$. Then we see that $a - a_1 \in J$ and $a - a_1 \in fA$. Since J is annihilated by a power of f we see that $a - a_1$ is a

nilpotent element. Hence $F^k(a - a_1)$ is zero for some large k. Thus after doing more replacements we get $a = a_1$.

We check condition (4). Let X, X', Z, E be as in Example 0EVH. By Lemma 0EVJ we see that

$$\mathcal{F}(X) = \operatorname{colim}_F \Gamma(X, \mathcal{O}_X) \longrightarrow \operatorname{colim}_F \Gamma(X', \mathcal{O}_{X'}) = \mathcal{F}(X')$$

is bijective. Since $E = \mathbf{P}^1_Z$ in this case we also see that $\mathcal{F}(Z) \to \mathcal{F}(E)$ is bijective. Thus the conclusion holds in this case as well. □